

**BIOGRAPHICAL SKETCH**

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NAME: Kelly A. Warren

eRA COMMONS USER NAME : KELLYWARREN

POSITION TITLE: Clinical Associate Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

| INSTITUTION AND LOCATION                | DEGREE<br>(if applicable) | Completion Date<br>MM/YYYY | FIELD OF STUDY            |
|-----------------------------------------|---------------------------|----------------------------|---------------------------|
| Pace University, Pleasantville, NY      | B.S.                      | 05/2003                    | Biology                   |
| Stony Brook University, Stony Brook, NY | M.S.                      | 5/2006                     | Biomedical Sciences       |
| Stony Brook University, Stony Brook, NY | Ph.D.                     | 12/2008                    | Physiology and Biophysics |

**A. Personal Statement**

I am excited to have put together an interdisciplinary team to explore outcomes of a physical exercise and virtual-reality mindfulness intervention in breast cancer patients. My education and research interests are within the biomedical sciences. I have demonstrated research productivity across physiologic disciplines including participation in studies evaluating respiratory neural control, neuroendocrinology of reproduction, and energy requirements during exercise. More recently, I serve as principal investigator for a study evaluating student health and wellness in first year graduate allied health students. I am also co-investigator on a project exploring physical fatigue in a population of patients with Lyme disease and participated on a service-learning project for health professional students to engage on interdisciplinary teams in an exercise-related intervention for children who are overweight or obese. For both activities, I have participated in data collection (e.g. submaximal exercise testing) in line with my role on the current proposed project. Additionally, my scholarly activity has provided me strengths across various research designs and methodologies, and I have experiences that include biomolecular bench work, *in vivo* and *in vitro* animal studies, and human subject research. In each of these research disciplines, I have demonstrated scholarly productivity through peer-reviewed research publications and presentations. I am eager to lead this collaborative team created to evaluate the role of wearable devices and mindfulness on breast cancer recovery, eventually contributing data to these measures in predicting comorbidities and extending to other populations.

**B. Positions, Scientific Appointments, and Honors**

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| 6/2003-8/2004  | MEDICAL RESEARCH ASSOCIATE, Brookhaven National Laboratory, Upton, NY     |
| 8/2004-12/2208 | GRADUATE RESEARCH ASSISTANT, Stony Brook University, Stony Brook, NY      |
| 1/2009-11/2013 | RESEARCH INSTRUCTOR, Stony Brook University, Stony Brook, NY              |
| 12/2013-8/2016 | RESEARCH ASSISTANT PROFESSOR, Stony Brook University, Stony Brook, NY     |
| 9/2016-8/2020  | CLINICAL RESEARCH PROFESSOR, Stony Brook University, Stony Brook, NY      |
| 9/2016-Present | RESEARCH ASSISTANT PROFESSOR (V), Stony Brook University, Stony Brook, NY |

9/2020-Present CLINICAL ASSOCIATE PROFESSOR, Stony Brook University, Stony Brook, NY

Other Experience and Professional Memberships:

2005-2016 Society for Neuroscience  
 2005-present American Physiological Society  
 2007-present Academy for the Advancement of Sciences

### C. Contributions to Science

1. My early training was in biomolecular bench work, *in vivo* and *in vitro* animal studies, where I studied the role of glutamatergic neurotransmitters on gasping in arterially perfused rats. This work, as well as other animal studies has influenced my work on human subjects looking at aerobic capacity in patients who are struggling with fatigue due to Lyme Disease.

a. **Warren, K.A.** and I.C. Solomon. Glutamatergic neurotransmission is not essential for, but plays a modulatory role in, the production of gasping in the arterially-perfused adult rat. *Adv. Exp. Med. Biol.* 2008. (Special Issue entitled "Modeling and Control of Breathing: Integration in Respiratory Control: From Genes to Systems").

b. **Warren, K.A.** and I.C. Solomon. Influence of chronic blockade of serotonin and norepinephrine reuptake mechanisms on the timing and patterning of eupnea and gasping in arterially-perfused adult mouse. *Respiratory Neurophysiology and Neurobiology*, 2012 Oct 15;184(1):9-15.

c. Simons, J and **Warren, KA.** Aerobic Capacity in Patients with Lyme Associated Fatigue. *International Lyme and Associated Diseases Society* (2023).

2. In addition to the above work, I have spent time both in basic sciences and more recently in human studies, exploring the relationship between hyperglycemia and hyperinsulinemia and health across several systems. This work has expanded to include the measurement of metabolic activity in several populations including amputees, young, obese children and college students who identify as being "stressed". This work positions me well to assist with the assessment and measurement of metabolic health in young adults.

a. A.L. Negrón, M. Beymer, G. Yu, **K.A. Warren**, M. Acosta-Martínez. Prolonged hyperglycemia & hyperinsulinemia increases BDNF mRNA expression in the posterior ventromedial hypothalamus and the dorsomedial hypothalamus of fed female rats. *Neuroscience*, September 2015 Volume 303: 422-432

b. Gandhi J, Dagur G, **Warren K**, Smith NL, Sheynkin YR, Zumbo A, Khan SA. The role of diabetes mellitus in sexual and reproductive health: an overview of pathogenesis, evaluation, and management. *Current Diabetes Reviews*, 13(6):573-581, 2017.

c. Gandhi J, Dagur G, **Warren K**, Smith NL, Khan SA. Genitourinary complications of diabetes mellitus: an overview of pathogenesis, evaluation, and management. *Current Diabetes Reviews*, 13(5):498-518, 2017.

d. Mayank Seth, **Kelly Warren**, Eric Lamberg. Minimal change in metabolic activity of individuals with trans- tibial amputation following prosthesis mass alterations. In review, *Journal of Prosthetics and Orthotics* (2020).

3. Lastly, my current work has shifted to include both physical and mental health in young adults, such as college students. Longitudinal effects of academic pressures and clinical rotations of health professions students have been examined across one year (2 semesters). Additionally, we have studied the effects of exercise and virtual reality on stress and cardiovascular health in college

students. Preliminary data reflects improvements in stress following both interventions. Data is currently being synthesized and includes mental health, body composition, metabolic markers, and cardiovascular measures.

- a. **Warren, K.** (2024). Longitudinal effects of graduate allied health education on physical and mental health. *International Journal of Exercise Science: Conference Proceedings*, 15(5), Article 21.
- b. **Warren, K.**, O'Brien, J., & Martino, S. (2024). *Interprofessional collaboration to advance the health and wellness of graduate students in allied health professional programs*. ASAHP, Atlanta, GA, USA.
- c. Martino, S.A., **Warren, K.**, Canterella, A., Morelli, P.J. (2024, March 6) *Pediatric Obesity and Long-term Cardiometabolic Outcomes: A Clinical Case Study*. Women in Medicine and Science Conference, Stony Brook University, Stony Brook, NY, USA.
- d. Martino, S.A., **Warren, K.A.**, Morelli, P.J., Zillo, C., & Zhang, J. (2023, November). *Pediatric Obesity and Cardiometabolic Health Outcomes – A Five-Year Journey*. Greater NY ACSM, New York, NY, USA.